

Module 3 - Database Integration - PostgreSQL, MySQL & MongoDB

Given data - Common Scenario: Student Course Registration System

HANDS-ON 1

Task 1: Create the Database and Tables

Step 1: Create Database

```
CREATE DATABASE college_db;
```

Use the database:

```
USE college_db;
```

Create departments Table

```
SELECT * from departments
```

Output

department_id	dept_name	hod_name	budget
1	Computer Science	Dr. Ramesh Kumar	850000
2	Electronics	Dr. Priya Nair	620000
3	Mechanical	Dr. Suresh Iyer	540000
4	Civil	Dr. Ananya Sharma	430000

SQL QUERY:

```
CREATE TABLE departments (
```

```
    department_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
dept_name VARCHAR(100) NOT NULL,  
hod_name VARCHAR(100),  
budget DECIMAL(12,2)  
);
```

Insert Sample Data

```
INSERT INTO departments (dept_name, hod_name,  
budget) VALUES  
  
('Computer Science', 'Dr. Ramesh Kumar', 850000.00),  
('Electronics', 'Dr. Priya Nair', 620000.00),  
('Mechanical', 'Dr. Suresh Iyer', 540000.00),  
('Civil', 'Dr. Ananya Sharma', 430000.00);
```

Create Students Table:

SQL:

```
CREATE TABLE students (  
    student_id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,
```

```
date_of_birth DATE,  
department_id INT,  
enrollment_year INT,  
FOREIGN KEY (department_id)  
REFERENCES departments(department_id)  
);
```

Insert Sample Data

```
INSERT INTO students
```

```
(first_name, last_name, email, date_of_birth,  
department_id, enrollment_year)
```

```
VALUES
```

```
('Arjun','Mehta','arjun.mehta@college.edu','2003-04-  
12',1,2022),
```

```
('Priya','Suresh','priya.suresh@college.edu','2003-07-  
25',1,2022),
```

```
('Rohan','Verma','rohan.verma@college.edu','2002-11-  
08',2,2021),
```

```
('Sneha','Patel','sneha.patel@college.edu','2004-01-  
30',3,2023),
```

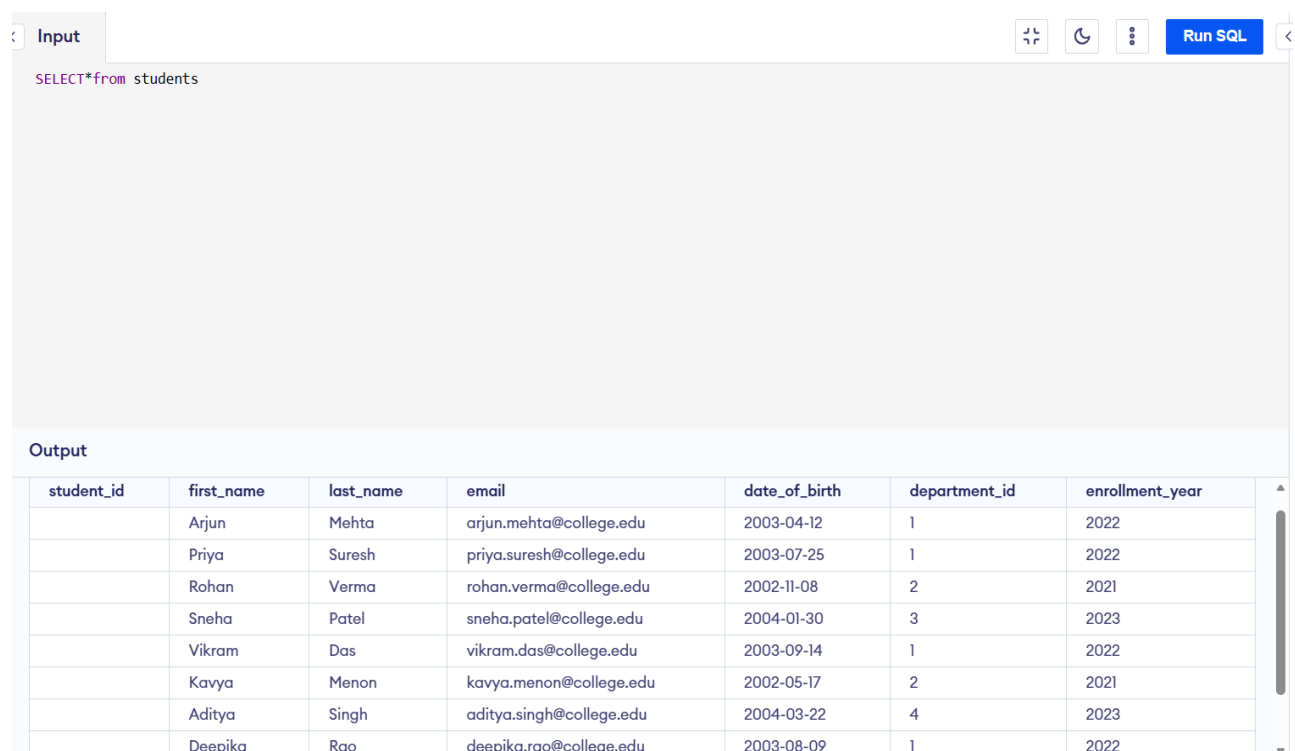
```
('Vikram','Das','vikram.das@college.edu','2003-09-  
14',1,2022),
```

('Kavya','Menon','kavya.menon@college.edu','2002-05-17',2,2021),

('Aditya','Singh','aditya.singh@college.edu','2004-03-22',4,2023),

('Deepika','Rao','deepika.rao@college.edu','2003-08-09',1,2022);

STUDENTS TABLE :



The screenshot shows a SQL query editor interface. At the top, there is a tab labeled 'Input' and a 'Run SQL' button. The query entered is 'SELECT * FROM students'. Below the query, the 'Output' section displays a table with 7 columns: student_id, first_name, last_name, email, date_of_birth, department_id, and enrollment_year. The table contains 8 rows of student data.

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year
	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022
	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022
	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021
	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021
	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022

CREATE COURSE TABLE:

SQL:

CREATE TABLE courses (

```
course_id INT AUTO_INCREMENT PRIMARY KEY,  
course_name VARCHAR(150) NOT NULL,  
course_code VARCHAR(20) UNIQUE,  
credits INT,  
department_id INT,  
FOREIGN KEY (department_id)  
REFERENCES departments(department_id)  
);
```

Insert sample data

```
INSERT INTO courses  
(course_name, course_code, credits, department_id)  
VALUES  
( 'Data Structures & Algorithms', 'CS101', 4, 1),  
( 'Database Management Systems', 'CS102', 3, 1),  
( 'Object Oriented Programming', 'CS103', 4, 1),  
( 'Circuit Theory', 'EC101', 3, 2),  
( 'Thermodynamics', 'ME101', 3, 3);
```

COURSE TABLE:

Input

Run SQL

```
SELECT*from courses
```

Output

course_id	course_name	course_code	credits	department_id
	Data Structures & Algorithms	CSI01	4	1
	Database Management Systems	CSI02	3	1
	Object Oriented Programming	CSI03	4	1
	Circuit Theory	ECI01	3	2
	Thermodynamics	MEI01	3	3

4. Enrollments

SQL

CREATE TABLE enrollments (

enrollment_id INT AUTO_INCREMENT PRIMARY KEY,

student_id INT,

course_id INT,

enrollment_date DATE,

grade CHAR(2),

FOREIGN KEY (student_id)

```
REFERENCES students(student_id),  
FOREIGN KEY (course_id)  
REFERENCES courses(course_id)  
);
```

ENROLLMENTS TABLE:

```
CREATE TABLE enrollments (  
    enrollment_id INT AUTO_INCREMENT PRIMARY KEY,  
    student_id INT,  
    course_id INT,  
    enrollment_date DATE,  
    grade CHAR(2),  
    FOREIGN KEY (student_id)  
        REFERENCES students(student_id),  
    FOREIGN KEY (course_id)  
        REFERENCES courses(course_id)  
);
```

Insert enrollments data:

```
INSERT INTO enrollments
```

```
(student_id, course_id, enrollment_date, grade)
```

```
VALUES
```

```
(1,1,'2022-07-01','A'),
```

```
(1,2,'2022-07-01','B'),
```

```
(2,1,'2022-07-01','B'),
```

```
(2,3,'2022-07-01','A'),
```

```
(3,4,'2021-07-01','A'),
```

```
(4,5,'2023-07-01',NULL),
```

```
(5,1,'2022-07-01','C'),
```

```
(5,2,'2022-07-01','A'),
```

```
(6,4,'2021-07-01','B'),
```

```
(7,5,'2023-07-01',NULL),
```

```
(8,1,'2022-07-01','A'),
```

```
(8,3,'2022-07-01','B');
```

ENROLLMENTS TABLE:

Input				
SELECT* FROM Enrollments				
Output				
enrollment_id	student_id	course_id	enrollment_date	grade
	1	1	2022-07-01	A
	1	2	2022-07-01	B
	2	1	2022-07-01	B
	2	3	2022-07-01	A
	3	4	2021-07-01	A
	4	5	2023-07-01	
	5	1	2022-07-01	C
	5	2	2022-07-01	A

5.PROFESSORS TABLE:

CREATE TABLE professors (

professor_id INT AUTO_INCREMENT PRIMARY KEY,

prof_name VARCHAR(100) NOT NULL,

email VARCHAR(100) UNIQUE,

department_id INT,

salary DECIMAL(10,2),

FOREIGN KEY (department_id)

REFERENCES departments(department_id)

);

Insert professors data

INSERT INTO professors

(prof_name, email, department_id, salary)

VALUES

('Dr. Anand Krishnan','anand.k@college.edu',1,95000.00),

('Dr. Meena Pillai','meena.p@college.edu',1,88000.00),

('Dr. Sunil Rajan','sunil.r@college.edu',2,82000.00),

('Dr. Latha Gopal','latha.g@college.edu',3,79000.00),

('Dr. Kartik Bose','kartik.b@college.edu',4,76000.00);

PROFESSORS TABLE:

```
SELECT * from professors
```

Output

professor_id	prof_name	email	department_id	salary
	Dr. Anand Krishnan	anand.k@college.edu	1	95000
	Dr. Meena Pillai	meena.p@college.edu	1	88000
	Dr. Sunil Rajan	sunil.r@college.edu	2	82000
	Dr. Latha Gopal	latha.g@college.edu	3	79000
	Dr. Kartik Bose	kartik.b@college.edu	4	76000

Expected Outcome: All 5 tables created with no errors.
DESCRIBE / \d commands show all columns and constraints.

TASK 2:

Expected Outcome

1NF compliance

2NF compliance

3NF compliance

EXAMPLE :

-- Task 2: Verify Normalisation

-- 1NF Analysis: -- All columns contain atomic values. --
Multiple phone numbers in one column would violate 1NF.

-- 2NF Analysis: -- Every non-key attribute depends on the entire primary key. -- In enrollments, grade and enrollment_date depend on both student_id and course_id.

-- 3NF Analysis: -- No transitive dependencies exist. --
Storing dept_name in students would violate 3NF. -- Hence the schema satisfies 3NF.

TASK 3: Alter and Extend the Schema

Goal: Use ALTER TABLE to safely extend an existing schema without data loss.

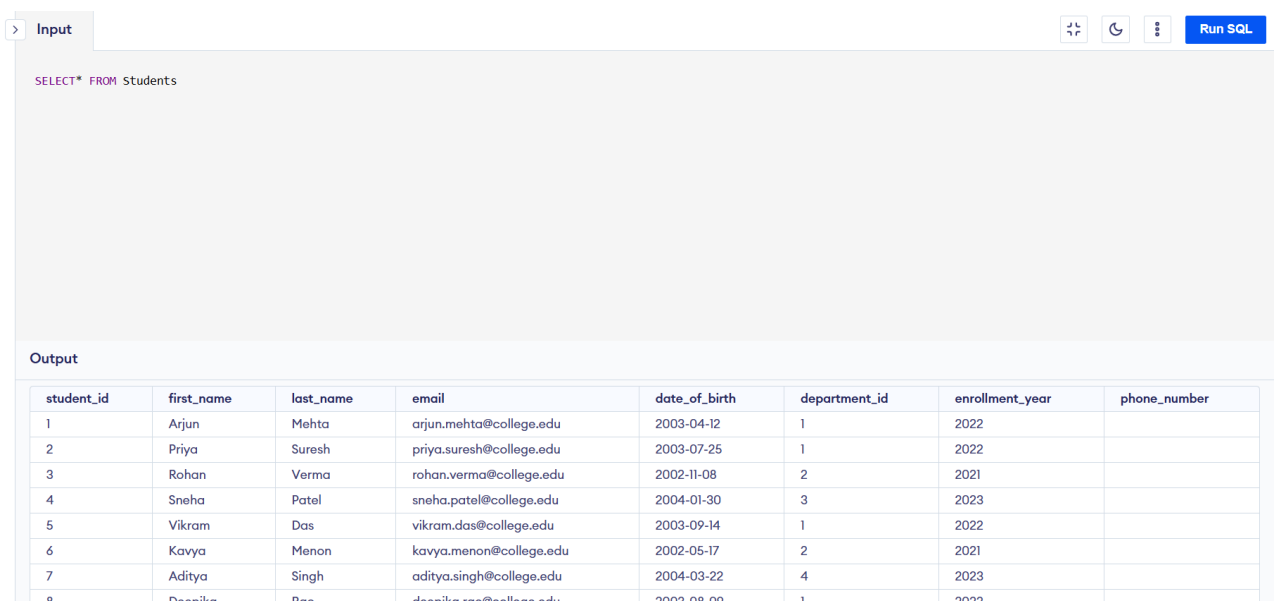
Step 10: Add phone_number to students

SQL

ALTER TABLE students

ADD phone_number VARCHAR(15);

EXAMPLE TABLE:



The screenshot shows a SQL IDE interface. At the top, there's a tab labeled 'Input' and a 'Run SQL' button. Below the input area, the query 'SELECT * FROM Students' is entered. The output area, labeled 'Output', displays a table with 8 rows and 8 columns. The columns are: student_id, first_name, last_name, email, date_of_birth, department_id, enrollment_year, and phone_number. The data in the table is as follows:

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year	phone_number
1	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022	
2	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022	
3	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021	
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023	
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022	
6	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021	
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023	
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022	

Step 11: Add max_seats to courses

SQL

ALTER TABLE courses

ADD max_seats INT DEFAULT 60;

EXAMPLE TABLE:

Input

SELECT * FROM Courses

Run SQL

Output

course_id	course_name	course_code	credits	department_id	max_seats
1	Data Structures & Algorithms	CSI01	4	1	60
2	Database Management Systems	CSI02	3	1	60
3	Object Oriented Programming	CSI03	4	1	60
4	Circuit Theory	ECI01	3	2	60
5	Thermodynamics	MEI01	3	3	60

Step 12: Add a CHECK constraint to enrollments

For MySQL 8+:

ALTER TABLE enrollments

ADD CHECK (grade IN ('A','B','C','D','F') OR grade IS NULL);

Step 13: Rename hod_name to head_of_dept

Input

SELECT * FROM Departments

Output

department_id	dept_name	head_of_dept	budget
1	Computer Science	Dr. Ramesh Kumar	850000
2	Electronics	Dr. Priya Nair	620000
3	Mechanical	Dr. Suresh Iyer	540000
4	Civil	Dr. Ananya Sharma	430000

Step 14: Drop the phone_number column (schema rollback)

SQL

ALTER TABLE students

DROP COLUMN phone_number;

EXAMPLE TABLE:

Input

Run SQL

SELECT * FROM Students

Output

student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year
1	Arjun	Mehta	arjun.mehta@college.edu	2003-04-12	1	2022
2	Priya	Suresh	priya.suresh@college.edu	2003-07-25	1	2022
3	Rohan	Verma	rohan.verma@college.edu	2002-11-08	2	2021
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023
5	Vikram	Das	vikram.das@college.edu	2003-09-14	1	2022
6	Kavya	Menon	kavya.menon@college.edu	2002-05-17	2	2021
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023
8	Deepika	Rao	deepika.rao@college.edu	2003-08-09	1	2022

